

D684

Introduction to Computer Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A cafeteria app contains this pseudocode: total $\leftarrow$ 0; FOR i $\leftarrow$ 1 TO 5: total $\leftarrow$ total + i; END FOR; PRINT total. The app was written by a student team during a weekend hackathon. What prints?

- A. 10
- B. 15
- C. 120
- D. 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Pseudocode: count $\leftarrow$ 0; x $\leftarrow$ 16; WHILE x > 1: x $\leftarrow$ x / 2; count $\leftarrow$ count + 1; END WHILE. What is count when the loop ends?

- A. 16
- B. 4
- C. 5
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A library system lets a patron borrow if (hasCard AND finesOwed < 10) OR isStaff. A patron has a card, owes \$12, is not staff, and has been a member for 20 years. Can the patron borrow?

- A. Yes, because of 20 years of membership.
- B. Yes, because the patron has a card.
- C. No.
- D. Yes, because only one condition needs to be true.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which input makes (A OR B) AND NOT C true?

- A. A = true, B = true, C = true
- B. A = false, B = true, C = false
- C. A = true, B = false, C = true
- D. A = false, B = false, C = false

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A network technician sees the 8-bit value 01101010 in a log entry from 3:14 a.m. on a router installed last year. What is its decimal value?

- A. 42
- B. 86
- C. 4
- D. 106

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which 8-bit binary number equals decimal 45?

- A. 10110100
- B. 01000101
- C. 00101100
- D. 00101101

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A contact list of 1,024 names is sorted alphabetically and stored on a phone with 8 GB of memory. Each binary-search comparison halves the names still in play. About how many comparisons does it take to narrow the list to one name?

- A. 512
- B. 1,024
- C. 10
- D. 32

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A binary search needs at most about 20 halvings to find any item. Roughly how many sorted items are in the list?

- A.** About 1,000,000
- B.** 20
- C.** 400
- D.** 40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A programmer meant to swap two values: $a \leftarrow 5$; $b \leftarrow 8$; $a \leftarrow b$; $b \leftarrow a$. The code was reviewed by two senior developers. What are the final values?

- A.** $a = 8$, $b = 8$
- B.** $a = 5$, $b = 8$, since reviewed code is correct
- C.** $a = 5$, $b = 5$
- D.** $a = 8$, $b = 5$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A laptop loses power in the middle of a software update. The user had an unsaved spreadsheet open and a report saved earlier to the SSD. The laptop is two years old and has 8 processor cores. Which data is lost?

- A.** The unsaved spreadsheet changes.
- B.** The report saved to the SSD.
- C.** It depends on the number of cores.
- D.** All data on the laptop.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student planning a school fundraiser splits the job into ticket sales, venue, food, and promotion, then works on each part separately. Last year's fundraiser raised \$2,300. Which computational-thinking skill is this?

- A.** Algorithm design.
- B.** Abstraction.
- C.** Pattern recognition.
- D.** Decomposition.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A city plans to use an algorithm trained on ten years of past arrest records to decide where to send police patrols. The vendor offers a low price and a fast install. What is the main ethical concern?

- A.** None; algorithms are objective.
- B.** Storing ten years of data costs too much.
- C.** Bias in past arrest patterns can be repeated and amplified, sending more patrols to the same areas.
- D.** The low price suggests poor quality.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A cafeteria app contains this pseudocode: $\text{total} \leftarrow 0$; FOR $i \leftarrow 1$ TO 5: $\text{total} \leftarrow \text{total} + i$; END FOR; PRINT total. ~~The app was written by a student team during a weekend hackathon.~~ What prints?

Target word: loop output **Decides:** The loop adds 1 through 5.

A	Dropped element	Drops the last pass (1 through 4).
B	Correct	$1 + 2 + 3 + 4 + 5 = 15$.
C	Inverted operation	Multiplies instead of adding.
D	Wrong stem number	The loop's end value, not the total.

2. Answer B Concept 1 • transfer

Pseudocode: $\text{count} \leftarrow 0$; $x \leftarrow 16$; WHILE $x > 1$: $x \leftarrow x / 2$; $\text{count} \leftarrow \text{count} + 1$; END WHILE. What is count when the loop ends?

Target word: count passes **Decides:** x goes $16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$.

A	Wrong stem number	The starting value of x .
B	Correct	Four halvings bring x to 1, and $1 > 1$ is false.
C	Adjacent category	Off by one: counts a pass for $x = 1$, which fails the test.
D	Dropped element	The value of x after one pass.

3. Answer C Concept 2

A library system lets a patron borrow if ($\text{hasCard AND finesOwed} < 10$) OR isStaff . A patron has a card, owes \$12, is not staff, ~~and has been a member for 20 years~~. Can the patron borrow?

Target word: evaluate the condition **Decides:** $\text{finesOwed} < 10$ is false, so the AND is false; isStaff is false.

A	Camouflage	Membership length is not in the condition.
B	Dropped element	Drops the fines condition inside the AND.
C	Correct	$(\text{true AND false}) \text{ OR false} = \text{false}$.
D	Inverted operation	Treats AND as OR.

4. Answer B Concept 2 • transfer

Which input makes $(A \text{ OR } B) \text{ AND NOT } C$ true?

Target word: find a true input — reversed **Decides:** Need at least one of A or B true and C false.

A	Dropped element	Drops the NOT; NOT true is false.
B	Correct	$(\text{false OR true}) \text{ AND true} = \text{true}$.
C	Adjacent category	The OR passes but $C = \text{true}$ fails NOT C.
D	Confusion set	The OR is false with no true input.

5. Answer D Concept 3

A network technician sees the 8-bit value 01101010 in a log entry ~~from 3:14 a.m. on a router installed last year~~. What is its decimal value?

Target word: binary to decimal **Decides:** 1s in the 64, 32, 8, and 2 places.

A	Dropped element	Drops the 64 bit.
B	Inverted operation	Reads the bits in reverse order (01010110).
C	Confusion set	Counts the 1s instead of adding place values.
D	Correct	$64 + 32 + 8 + 2 = 106$.

6. Answer D Concept 3 • transfer

Which 8-bit binary number equals decimal 45?

Target word: decimal to binary **Decides:** $45 = 32 + 8 + 4 + 1$.

A	Inverted operation	The correct bits written in reverse (180).
B	Confusion set	Encodes the digits 4 and 5 separately, not the number 45.
C	Dropped element	Missing the 1 bit (44).
D	Correct	$32 + 8 + 4 + 1 = 45$.

7. Answer C Concept 4

A contact list of 1,024 names is sorted alphabetically ~~and stored on a phone with 8 GB of memory~~. Each binary-search comparison halves the names still in play. About how many comparisons does it take to narrow the list to one name?

Target word: binary search steps **Decides:** $2^{10} = 1,024$.

A	Confusion set	The average for linear search.
B	Adjacent category	The worst case for linear search.
C	Correct	1,024 halves to 1 in 10 steps.
D	Wrong stem number	The square root of 1,024, not the halving count.

8. Answer A Concept 4 • transfer

A binary search needs at most about 20 halvings to find any item. Roughly how many sorted items are in the list?

Target word: solve for list size **Decides:** $n \approx 2^{20}$.

A	Correct	$2^{20} = 1,048,576$.
B	Wrong stem number	The number of halvings.
C	Inverted operation	Squares 20 instead of raising 2 to the 20th.
D	Confusion set	Doubles 20 once.

9. Answer A Concept 5

A programmer meant to swap two values: $a \leftarrow 5$; $b \leftarrow 8$; $a \leftarrow b$; $b \leftarrow a$. ~~The code was reviewed by two senior developers.~~ What are the final values?

Target word: trace assignments **Decides:** After $a \leftarrow b$, the original 5 is lost.

A	Correct	a becomes 8, then b copies a's new value, 8.
B	Camouflage	Review does not change what the code does.
C	Inverted operation	Runs the copies in the wrong direction.
D	Adjacent category	The intended swap, which needs a temporary variable.

10. Answer A Concept 6

A laptop loses power in the middle of a software update. The user had an unsaved spreadsheet open and a report saved earlier to the SSD. ~~The laptop is two years old and has 8 processor cores.~~ Which data is lost?

Target word: what is lost **Decides:** Unsaved work lives only in RAM.

A	Correct	RAM is volatile; the SSD file persists.
B	Inverted operation	SSD storage is non-volatile.
C	Camouflage	Core count does not decide volatility.
D	Absolute language	Saved files survive a power loss.

11. Answer D Concept 7

A student planning a school fundraiser splits the job into ticket sales, venue, food, and promotion, then works on each part separately. ~~Last year's fundraiser raised \$2,300.~~ Which computational-thinking skill is this?

Target word: name the skill **Decides:** Breaking a big problem into parts.

A	Adjacent category	Writing ordered steps, which comes after the parts are known.
B	Confusion set	Hiding unneeded detail, not splitting.
C	Confusion set	Spotting similarities across problems.
D	Correct	Splitting a problem into smaller parts.

12. Answer C Concept 8

A city plans to use an algorithm trained on ten years of past arrest records to decide where to send police patrols. ~~The vendor offers a low price and a fast install.~~ What is the main ethical concern?

Target word: ethical risk **Decides:** Training data reflects past human decisions.

A	Absolute language	They learn whatever patterns, fair or not, the data holds.
B	Adjacent category	A cost issue, not an ethical one.
C	Correct	A feedback loop can entrench historical bias.
D	Camouflage	Price does not decide the ethical issue.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Tracing loops	Step through a loop one pass at a time, updating each variable and checking the condition before every pass.	Off-by-one: counting the pass that fails the test, or stopping one early.	Pseudocode with FOR or WHILE and “what prints?”
2	Boolean logic	AND needs both sides true; OR needs at least one; NOT flips; parentheses are evaluated first.	Reading AND as OR, or skipping the NOT.	A compound condition with AND, OR, NOT.
3	Binary place values	Each bit from the right is worth 1, 2, 4, 8, 16, 32, 64, 128; add the values where the bit is 1.	Reading bits left to right as ones place, or counting the 1s.	An 8-bit string and “decimal value.”
4	Binary search halves the work	On sorted data, each comparison halves the remaining items, so about $\log_2(n)$ comparisons are needed.	Using linear-search counts.	A sorted list size that is a power of 2.
5	Sequential assignment	Statements run in order; once a variable is overwritten, its old value is gone unless saved in a temporary variable.	Assuming a two-line swap works.	$a \leftarrow b$ followed by $b \leftarrow a$.
6	Volatile vs. non-volatile storage	RAM is fast but loses its contents without power; SSDs and hard drives keep data.	Thinking saved files are lost or unsaved work survives.	A power loss with saved and unsaved data.
7	Computational thinking	Decomposition breaks a problem into parts; pattern recognition spots similarities; abstraction hides unneeded detail; algorithms give step-by-step solutions.	Mixing up the four names.	“Breaks the job into...” or “ignores details like...”
8	Algorithmic bias and ethics	Algorithms trained on biased historical data can reproduce and amplify that bias; outputs are not automatically objective.	Believing algorithms are neutral by nature.	A model trained on past human decisions about people.

Course scope (public description): Introduction to Computer Science introduces learners to the field of computer science and its essential concepts. In this course students will explore, among other topics, programming basics through language-agnostic pseudocode, computational thinking and problem-solving, algorithms, hardware, and social and ethical considerations in the field. This course is designed for beginners to gain a solid understanding of computer science principles and their applications and prepare students for further study in the field. There are no prerequisites for this course.